

Question 04. Aryan worked on the farm with his father.

He saw that his father added earthworms to the soil. He came to his father and told him not to do so. His father explained why he was doing so and Aryan was satisfied with the explanation.

(i) Why were earthworms added to the soil?

(ii) What values do you observe in Aryan's father?

Answer: (i) Earthworms are detritivores which cause the breakdown of detritus into small, simpler forms. This helps in improving soil quality.

(ii) Aryan's father is intelligent, hard worker and environment friendly.

Question 05. Suppose, you went to a zoological park with your family? And one of member of your family curious about that the why animals are in the net or cage or birdcage? How can you explain the biodiversity and its in-situ conservation to your family member?

Answer: Biodiversity: The variety of plant and animal life in the world or in a particular habitat, a high level of which is usually considered to be important and desirable.

Or

The variety of life on the earth including plant, animals and microorganism.

Or

Biodiversity is the variation among living organisms from different sources including terrestrial, marine and desert ecosystems, and the ecological complexes of which they are a part.

In-situ conservation of bio-diversity: It is the methods of conserving all the living species, especially the wild and endangered species in their natural habitats and environment. In-situ conservation of Biodiversity includes biosphere reserves, national parks, wildlife sanctuaries, etc.

Some important point about this:

- (i) Requires large populations
- (ii) Need large area
- (iii) Less expensive
- (iv) Greater mobility to the organism because of large area

Question 08. Mention the role of healthy ecosystem services as a prerequisite for a wide range of economic, environmental and aesthetic goods and services.

Answer: Ecosystem services are the products of ecosystem processes. Forests are the major source of ecosystem services and are prerequisite for environmental, aesthetic goods and indirect economic values in the following ways:

- (i) Carbon-fixation huge amount of CO₂ in the atmosphere is removed naturally and fixed by plants into organic molecules and energy through photosynthesis. All the other trophic levels, i.e. consumers depend upon this energy produced by them.
- (ii) Release of oxygen by the producers as a byproduct in the process of photosynthesis, improves the air quality and supports life on earth.
- (iii) Soil formation and soil protection are the major ecosystem services accounting for nearly 50% of their total worth.
- (iv) Plant cover protects the soil from drastic changes in the temperature. There is little wind or water erosion as soil particles are not exposed to them.
- (v) The soil remains spongy and fertile. There occurs no landslides and floods.
- (vi) Nutrient cycling There is no depletion of nutrients, but the same are repeatedly circulated.